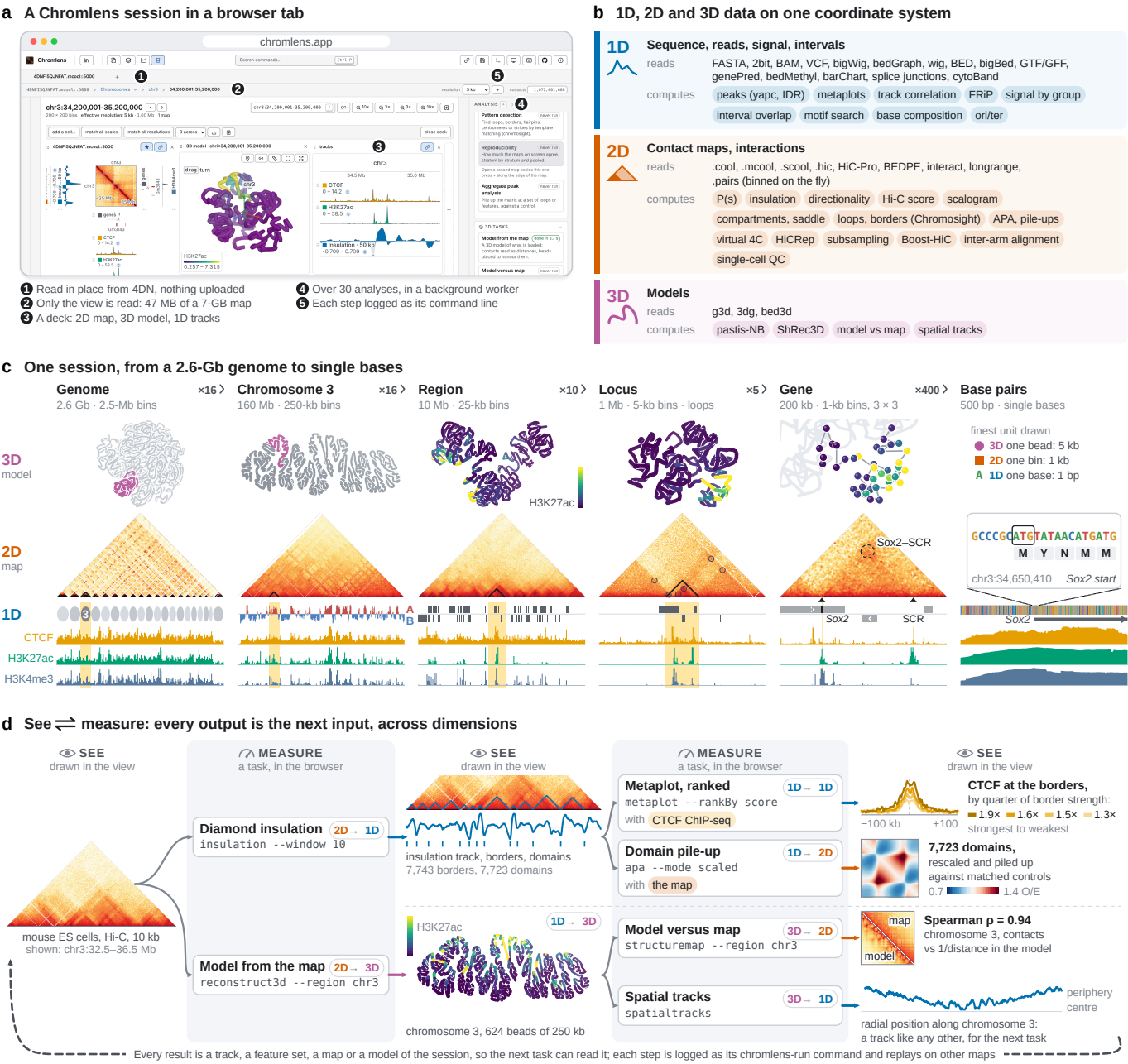

a A Chromlens session in a browser tab
chromlens.app
1 Read in place from 4DN, nothing uploaded
2 Only the view is read: 47 MB of a 7-GB map
3 A deck: 2D map, 3D model, 1D tracks
4 Over 30 analyses, in a background worker
5 Each step logged as its command line
b 1D, 2D and 3D data on one coordinate system
1D
Sequence, reads, signal, intervals
reads
FASTA, 2bit, BAM, VCF, bigWig, bedGraph, wig, BED, bigBed, GTF/GFF, genePred, bedMethyl, barChart, splice junctions, cytoBand
computes
peaks (yapc, IDR)
metaplots
track correlation
FRiP
signal by group
interval overlap
motif search
base composition
ori/ter
2D
Contact maps, interactions
reads
.cool, .mcool, .scool, .hic, HiC-Pro, BEDPE, interact, longrange, .pairs (binned on the fly)
computes
P(s)
insulation
directionality
Hi-C score
scalogram
compartments, saddle
loops, borders (Chromosight)
APA, pile-ups
virtual 4C
HiCRep
subsampling
Boost-HiC
inter-arm alignment
single-cell QC
3D
Models
reads
g3d, 3dg, bed3d
computes
pastis-NB
ShRec3D
model vs map
spatial tracks
c One session, from a 2.6-Gb genome to single bases
Genome
×16
2.6 Gb · 2.5-Mb bins
Chromosome 3
×16
160 Mb · 250-kb bins
Region
×10
10 Mb · 25-kb bins
Locus
×5
1 Mb · 5-kb bins · loops
Gene
×400
200 kb · 1-kb bins, 3 × 3
Base pairs
500 bp · single bases
3D model
2D map
1D
finest unit drawn
3D one bead: 5 kb
2D one bin: 1 kb
1D one base: 1 bp
H3K27ac
Sox2–SCR
Sox2
SCR
chr3:34,650,410
Sox2 start
CTCF
H3K27ac
H3K4me3
d See ⇌ measure: every output is the next input, across dimensions
SEE
drawn in the view
MEASURE
a task, in the browser
mouse ES cells, Hi-C, 10 kb
shown: chr3:32.5–36.5 Mb
Diamond insulation
2D → 1D
insulation --window 10
insulation track, borders, domains
7,743 borders, 7,723 domains
Model from the map
2D → 3D
reconstruct3d --region chr3
chromosome 3, 624 beads of 250 kb
Metaplot, ranked
1D → 1D
metaplot --rankBy score
with CTCF ChIP-seq
−100 kb
+100
CTCF at the borders,
by quarter of border strength:
1.9× 1.6× 1.5× 1.3×
strongest to weakest
Domain pile-up
1D → 2D
apa --mode scaled
with the map
7,723 domains,
rescaled and piled up
against matched controls
0.7
1.4 O/E
Model versus map
3D → 2D
structuremap --region chr3
map
model
Spearman ρ = 0.94
chromosome 3, contacts
vs 1/distance in the model
Spatial tracks
3D → 1D
spatialtracks
periphery
centre
radial position along chromosome 3:
a track like any other, for the next task
Every result is a track, a feature set, a map or a model of the session, so the next task can read it; each step is logged as its chromlens-run command and replays on other maps